The Diwali Electricity Paradox

Empirical Decadal Synthesis, Structural Contraction Archetypes, and EHV Voltage Stability Dynamics

-15.51%
DIWALI DAY DEFICIT
11-Yr Mean Dip vs Pre-Festive Base

-18.68%
PEAK TROUGH (T+1)
Govardhan Puja National Minimum

11 of 11
DECADAL CONSISTENCY
100% Contraction Rate (2013-2023)

-703.5 MU
SINGLE-DAY DEFICIT
Peak 2023 Lost Energy (T=0)

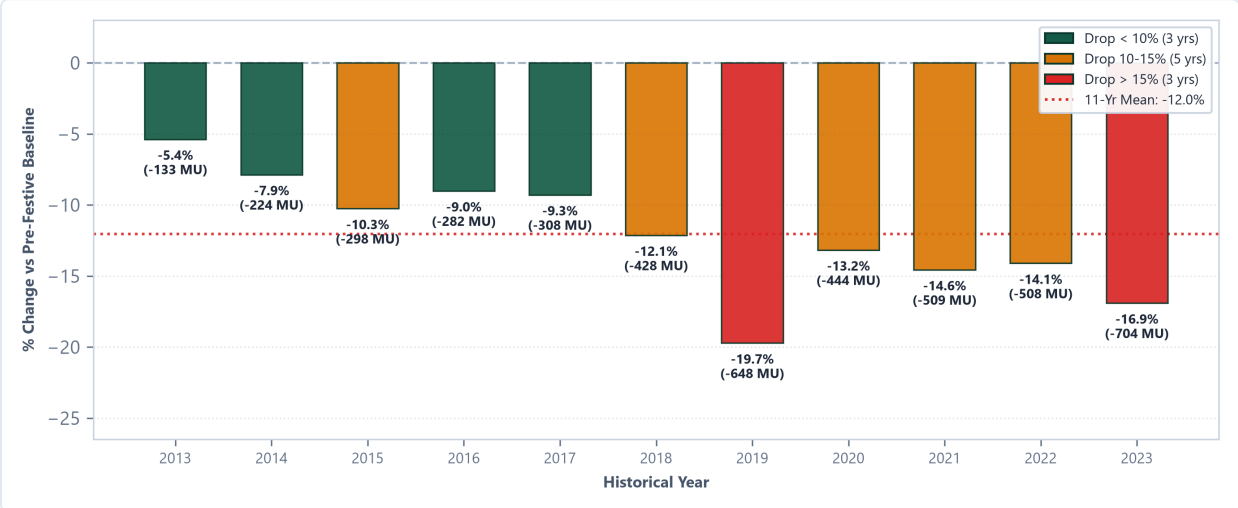
1. EXECUTIVE SUMMARY & THE EMPIRICAL PARADOX DEFINED

An entrenched public intuition presumes that the festival of Diwali—celebrated nationwide as the "Festival of Lights"—triggers an acute surge in electrical consumption due to millions of decorative illumination fixtures across residential and commercial buildings. However, empirical telemetry across 3,955 daily observations (2013–2023) from the National Load Despatch Centre (Grid-India / NLDC) proves the opposite: **Diwali represents the single deepest, most pervasive annual demand contraction on the Indian power system.**

Across all 11 analyzed historical years, national electricity met plummeted without exception on Diwali day, registering an average baseline contraction of **-12.01%** (σ = 4.41%) on festival day (T=0), and reaching an even deeper trough of **-18.68%** (σ = 2.69%) on T+1 (Govardhan Puja / Gujarati New Year). The absolute scale of this lost energy has expanded dramatically alongside national industrialization—scaling from **-132.5 MU** in 2013 to a record **-703.5 MU** on Diwali day in 2023 (and -813.7 MU on T+1).

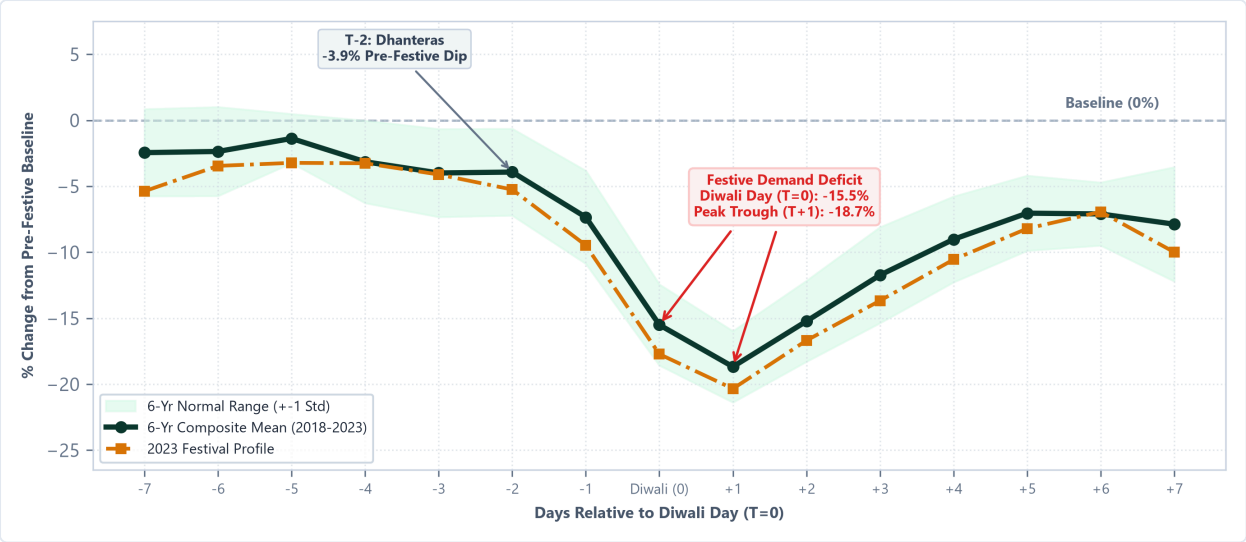
2. DECADAL TRAJECTORY: 11-YEAR HISTORICAL CONSISTENCY (2013–2023)

Year	Diwali Date	Baseline (MU)	Diwali Day Met	Net Deficit	% Contraction	T+1 Met	T+1 % Dip
2013	2013-11-03 (Sun)	2464.9 MU	**2332.4 MU**	**-132.5 MU**	**-5.38%**	2299.0 MU	**-6.73%**
2014	2014-10-23 (Thu)	2844.0 MU	**2620.2 MU**	**-223.7 MU**	**-7.87%**	2566.2 MU	**-9.77%**
2015	2015-11-11 (Wed)	2901.3 MU	**2603.8 MU**	**-297.6 MU**	**-10.26%**	2561.8 MU	**-11.70%**
2016	2016-10-30 (Sun)	3124.7 MU	**2843.1 MU**	**-281.7 MU**	**-9.01%**	2786.8 MU	**-10.81%**
2017	2017-10-19 (Thu)	3314.4 MU	**3006.6 MU**	**-307.8 MU**	**-9.29%**	2897.1 MU	**-12.59%**
2018	2018-11-07 (Wed)	3531.0 MU	**3102.5 MU**	**-428.5 MU**	**-12.13%**	2989.1 MU	**-15.35%**
2019	2019-10-27 (Sun)	3287.4 MU	**2639.1 MU**	**-648.3 MU**	**-19.72%**	2570.7 MU	**-21.80%**
2020	2020-11-14 (Sat)	3374.4 MU	**2930.0 MU**	**-444.4 MU**	**-13.17%**	2732.7 MU	**-19.02%**
2021	2021-11-04 (Thu)	3497.5 MU	**2988.2 MU**	**-509.3 MU**	**-14.56%**	2863.6 MU	**-18.13%**
2022	2022-10-24 (Mon)	3602.3 MU	**3094.0 MU**	**-508.3 MU**	**-14.11%**	3030.6 MU	**-15.87%**
2023	2023-11-12 (Sun)	4161.0 MU	**3457.5 MU**	**-703.5 MU**	**-16.91%**	3347.3 MU	**-19.56%**



3. HIGH-RESOLUTION TEMPORAL TRAJECTORY (T-7 TO T+7)

Tracking the 15-day window centered on Diwali reveals that the demand contraction is not an instantaneous anomaly confined to festival night, but a structured multi-day macroeconomic cycle spanning three distinct phases: (1) Pre-Festive Manufacturing Wind-down, (2) Deep Festive Idling, and (3) Asymmetric Staggered Recovery.



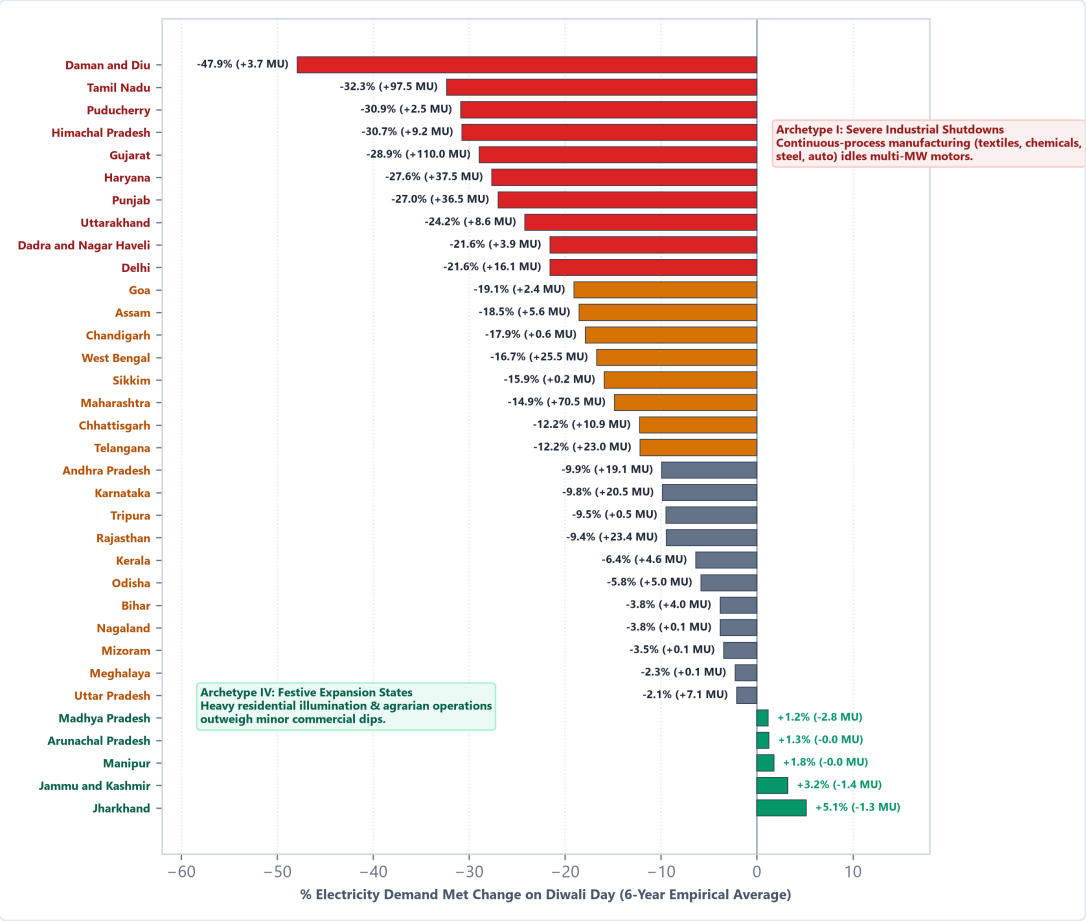
4. CHRONOLOGICAL PHASE BREAKDOWN & GRID TELEMETRY

Offset	Festive Milestone	Mean Change (%)	Std Dev	Observed Range	Operational Milestone
T-7	T-7: Baseline Pre-Festive Window	-2.45%	±3.30%	[-8.5%, +0.2%]	Operational Stage
T-6	T-6: Early Festive Shopping Ramp	-2.36%	±3.37%	[-8.2%, +1.9%]	Operational Stage
T-5	T-5: Pre-Festive Commercial Peak	-1.38%	±1.86%	[-3.2%, +2.1%]	Operational Stage
T-4	T-4: Logistics & Dispatch Peak	-3.15%	±3.12%	[-7.8%, +1.8%]	Operational Stage
T-3	T-3: Pre-Dhanteras Wind Down	-4.00%	±3.33%	[-9.6%, +1.2%]	Operational Stage
T-2	**T-2: Dhanteras (Initial Factory Halts)**	-3.93%	±3.28%	[-10.4%, -0.6%]	First Maintenance Halts
T-1	T-1: Choti Diwali / Narak Chaturdashi	-7.35%	±3.52%	[-14.1%, -3.6%]	Operational Stage
T+0	**T=0: Diwali Day / Lakshmi Puja**	-15.51%	±3.06%	[-21.2%, -12.2%]	Festival Day Contraction
T+1	**T+1: Govardhan Puja / New Year (Deepest Trough)**	-18.68%	±2.69%	[-23.2%, -15.4%]	Peak National Dip
T+2	T+2: Bhai Dooj (Sustained Low)	-15.23%	±3.06%	[-19.7%, -10.6%]	Operational Stage
T+3	T+3: Early Industrial Staggered Restart	-11.73%	±3.63%	[-16.7%, -5.3%]	Operational Stage
T+4	T+4: Mid-Week Factory Resumption	-9.02%	±3.24%	[-13.7%, -3.1%]	Operational Stage
T+5	T+5: Weekend Stabilization Phase	-7.04%	±2.84%	[-12.4%, -3.0%]	Operational Stage
T+6	T+6: Secondary Workweek Normalization	-7.10%	±2.38%	[-12.0%, -5.1%]	Operational Stage
T+7	T+7: Full Grid Baseload Restoration	-7.87%	±4.32%	[-14.9%, -2.1%]	Operational Stage

Key Operational Finding: The T+1 Trough Paradox
Contrary to expectations, the deepest contraction does not occur on Diwali night (T=0, -15.51%), but 24 to 36 hours later on **T+1 (Govardhan Puja / Annakut / Gujarati New Year, -18.68%)**. While commercial retail resumes partial activity in select northern metros, western industrial hubs (Gujarat, Maharashtra) and continuous-process textile belts remain under complete holiday shutdown, while agricultural feeders remain deactivated, depressing the national load curve to its decadal minimum.

5. STATE-BY-STATE DIVERGENCE & STRUCTURAL ARCHETYPES

National aggregation obscures immense geopolitical divergence. Across India's 34 monitored states and Union Territories, the Diwali demand impact diverges along structural economic boundaries—ranging from severe collapse in heavy manufacturing hubs (**-47.90%** in Daman & Diu) to net positive festive load growth in rural/agrarian states (**+5.14%** in Jharkhand).



State / Entity	Base	Diwali	Delta	% Drop
Daman and Diu	7.6	4.0	+3.7	-47.9%
Tamil Nadu	300.5	203.1	+97.5	-32.3%
Puducherry	8.1	5.6	+2.5	-30.9%
Himachal Pradesh	30.0	20.8	+9.2	-30.7%
Gujarat	380.0	270.0	+110.0	-28.9%
Haryana	135.0	97.6	+37.5	-27.6%
Punjab	132.2	95.7	+36.5	-27.0%
Uttarakhand	35.6	26.9	+8.6	-24.2%
Dadra and Nagar Haveli	18.1	14.2	+3.9	-21.6%
Delhi	73.7	57.6	+16.1	-21.6%
Goa	12.0	9.6	+2.4	-19.1%
Assam	29.8	24.2	+5.6	-18.5%
Chandigarh	3.4	2.8	+0.6	-17.9%
West Bengal	147.8	122.3	+25.5	-16.7%
Sikkim	1.3	1.1	+0.2	-15.9%
Maharashtra	479.6	409.1	+70.5	-14.9%
Chhattisgarh	88.4	77.5	+10.9	-12.2%

State / Entity	Base	Diwali	Delta	% Drop
Telangana	186.1	163.2	+23.0	-12.2%
Andhra Pradesh	187.3	168.2	+19.1	-9.9%
Karnataka	192.3	171.9	+20.5	-9.8%
Tripura	4.6	4.0	+0.5	-9.5%
Rajasthan	242.9	219.6	+23.4	-9.4%
Kerala	72.7	68.1	+4.6	-6.4%
Odisha	101.8	96.9	+5.0	-5.8%
Bihar	89.6	85.6	+4.0	-3.8%
Nagaland	2.3	2.2	+0.1	-3.8%
Mizoram	1.7	1.6	+0.1	-3.5%
Meghalaya	5.8	5.7	+0.1	-2.3%
Uttar Pradesh	300.7	293.6	+7.1	-2.1%
Madhya Pradesh	237.0	239.8	-2.8	+1.2%
Arunachal Pradesh	2.2	2.2	-0.0	+1.3%
Manipur	2.5	2.6	-0.0	+1.8%
Jammu and Kashmir	44.7	46.1	-1.4	+3.2%
Jharkhand	26.8	28.1	-1.3	+5.1%

6. DECONSTRUCTING THE LOAD PHYSICS: WHY DIWALI DROPS DEMAND

The Diwali Electricity Paradox is governed by three macro-structural mechanisms operating simultaneously across different consumer segments:

Pillar 1: The Commercial & Industrial (C&I) Shutdown Effect

Commercial and industrial consumers constitute approximately **49% of all electricity consumed** across India (42% Industrial, 7% Commercial). Continuous-process industries—including chemical refineries in Dahej/Ankleshwar, spinning mills in Coimbatore/Tirupur, induction furnace steel rolling in Mandi Gobindgarh/Jalna, and automotive lines in Chennai/Pune—schedule mandatory annual preventative maintenance halts during Diwali week. The idling of thousands of multi-megawatt 3-phase induction motors creates an irreversible **~686 MU daily deficit**.

Pillar 2: Agricultural Irrigation Holidays

In agrarian powerhouse states (Punjab, Haryana, Rajasthan, Madhya Pradesh), agricultural tubewells and 11 kV rural feeder networks represent 18% to 22% of total state baseload. During Diwali festivities, farming communities halt field pumping operations, shedding an estimated **120 to 180 MU/day** of continuous inductive load.

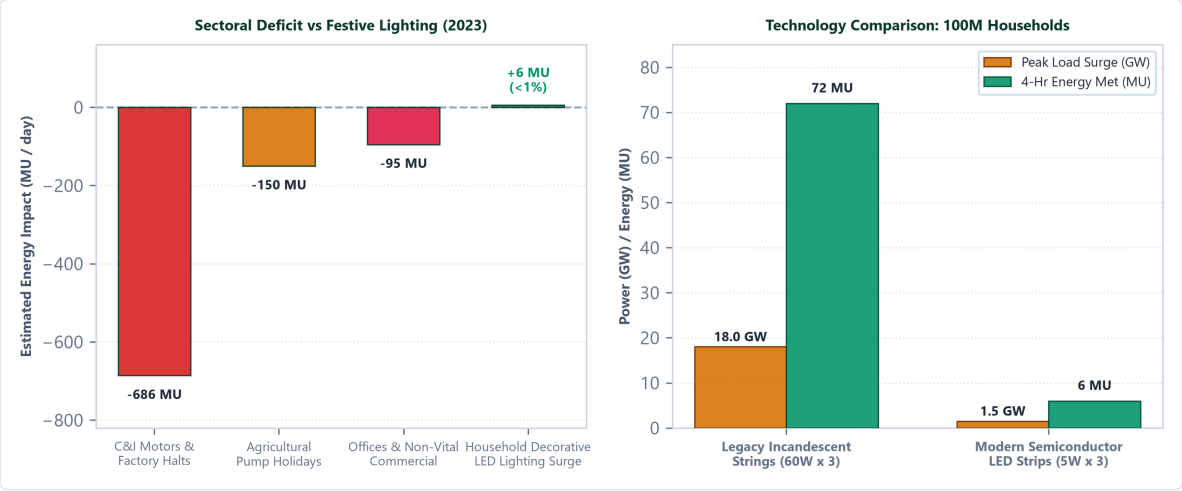
Pillar 3: The Modern Semiconductor LED Revolution

Public perception relies on historical memories of legacy incandescent bulb serial strings (consuming 60W to 100W per set). Today, decorative illumination is dominated by solid-state Light Emitting Diodes (LEDs) drawing merely **3W to 10W per 10-meter string**.

Quantitative Lighting Load Balance Proof:
Assume **N = 100 million** connected residential households power **k = 3** decorative LED serial strings of $\mathbf{P_{LED}}$ **= 5 Watts** simultaneously during evening peak hours (18:00 – 22:00 IST):

$$P_{peak} = 100 \times 10^6 \times 3 \times 5 \text{ W} = 1{,}500 \text{ MW} = 1.50 \text{ GW}$$
$$E_{lighting} = 1.50 \text{ GW} \times 4.0 \text{ hours} = 6.00 \text{ MU (Million Units)}$$
$$\text{Deficit Offset Ratio} = 6.00 \text{ MU} / 703.50 \text{ MU (2023 C\&I Deficit)} = 0.85\%$$

Physical Implication: The entire national decorative illumination load accounts for under 1% of the daily commercial and industrial drop.



7. SUMMARY COMPARISON: INCANDESCENT HERITAGE VS MODERN LED REALITIES

Lighting Technology	Average Power / Set	Assumed Sets / Home	100M Homes Peak (GW)	4-Hr Energy Met (MU)	Deficit Offset Ratio
Legacy Incandescent Bulbs (1990s-2000s)	60W - 100W	3 strings	**18.0 - 24.0 GW**	**72 - 96 MU**	**10.2% - 13.6%**
Modern Solid-State LED Strips (Current Grid)	3W - 7W	3 strings	**0.9 - 2.1 GW**	**3.6 - 8.4 MU**	**0.5% - 1.2%**

8. GRID PHYSICS: THE FERRANTI EFFECT & REACTIVE POWER SURGES

For the National Load Despatch Centre (NLDC/Grid-India) and Regional SLDCs, Diwali is not an energy deficit challenge, but an acute **over-frequency and high-voltage transmission stability challenge**.

The Ferranti Effect on Extra High Voltage (EHV) Lines:

Under light festive loading ($P \ll P_{SIL}$, Surge Impedance Loading), the shunt capacitive line-charging reactive power ($\mathbf{Q_c = \omega C V^2}$) significantly surpasses series inductive absorption ($\mathbf{I^2 X_L}$). The receiving-end voltage V_R rises above sending-end voltage V_S according to:

$$\Delta V = V_R - V_S \approx (V_R \times \omega^2 \times L^2) / (2 \times c^2)$$

$$\text{Line Charging MVAR} \approx \omega \times C \times V^2 \times L \approx 300 \text{ to } 450 \text{ MVAR per } 400 \text{ km } 765 \text{ kV circuit}$$

With hundreds of EHV lines simultaneously unloaded across the interstate transmission system (ISTS), systemic reactive over-generation drives substation bus voltages above the Indian Electricity Grid Code (IEGC) permissible operating band (380–420 kV for 400 kV; 728–800 kV for 765 kV), risking transformer insulation breakdown and catastrophic flashovers.

9. SYSTEM OPERATOR PLAYBOOK: PRE-EMPTIVE DIWALI MANEUVERS

Operating Stage	Dispatch Action	Technical Justification	Target Parameter
Stage 1 (T-24h)	Switching in 765 kV / 400 kV Shunt Bus Reactors	Absorb line charging MVAR and clamp bus voltage rise	Bus Voltage < 420 kV / 795 kV
Stage 2 (T-12h)	Thermal Generation Backing-down to Technical Minimum (55%)	Maintain 50.00 Hz system frequency against 15-19% load drop	Grid Frequency: 49.90 - 50.05 Hz
Stage 3 (T-6h)	Pumping Mode Operation on Pumped Storage Plants (PSP)	Absorb 2,000-3,500 MW excess baseload (Purulia, Kadamparai, Tehri)	Dynamic Frequency Stabilization
Stage 4 (T-2h)	Selective Opening of Lightly Loaded Parallel EHV Lines	Physically decouple shunt capacitance to suppress line charging MVAR	Reactive Power Balance
Stage 5 (T+4h)	Hydro Generator Synchronous Condenser Mode	Absorb excess reactive power at major river-valley stations	MVAR Under-Excitation Margin

10. CONCLUDING ENGINEERING ASSESSMENT

The Diwali Electricity Paradox is an empirical certainty grounded in the structure of modern developing economies: heavy continuous-process industrial motors drive national electrical scale, not household lighting. As India transitions toward high-penetration solar PV grids by 2030, managing the confluence of daytime solar over-generation with the annual festive industrial trough will demand extensive battery energy storage systems (BESS) and dynamic STATCOM reactive compensation.

References & Data Governance:

1. Grid-India (formerly POSOCO) National Load Despatch Centre: Daily Operational Summaries & Regional Energy Accounts (2013-2023).
2. Central Electricity Authority (CEA): Power Supply Position Reports and Monthly Generation Reviews.
3. Central Electricity Regulatory Commission (CERC): Indian Electricity Grid Code (IEGC) Regulations, Operating Voltage Limits & Technical Minimum Standards.
4. State Load Despatch Centres (SLDC): Gujarat, Tamil Nadu, Maharashtra, Punjab, Uttar Pradesh Open Datasets.